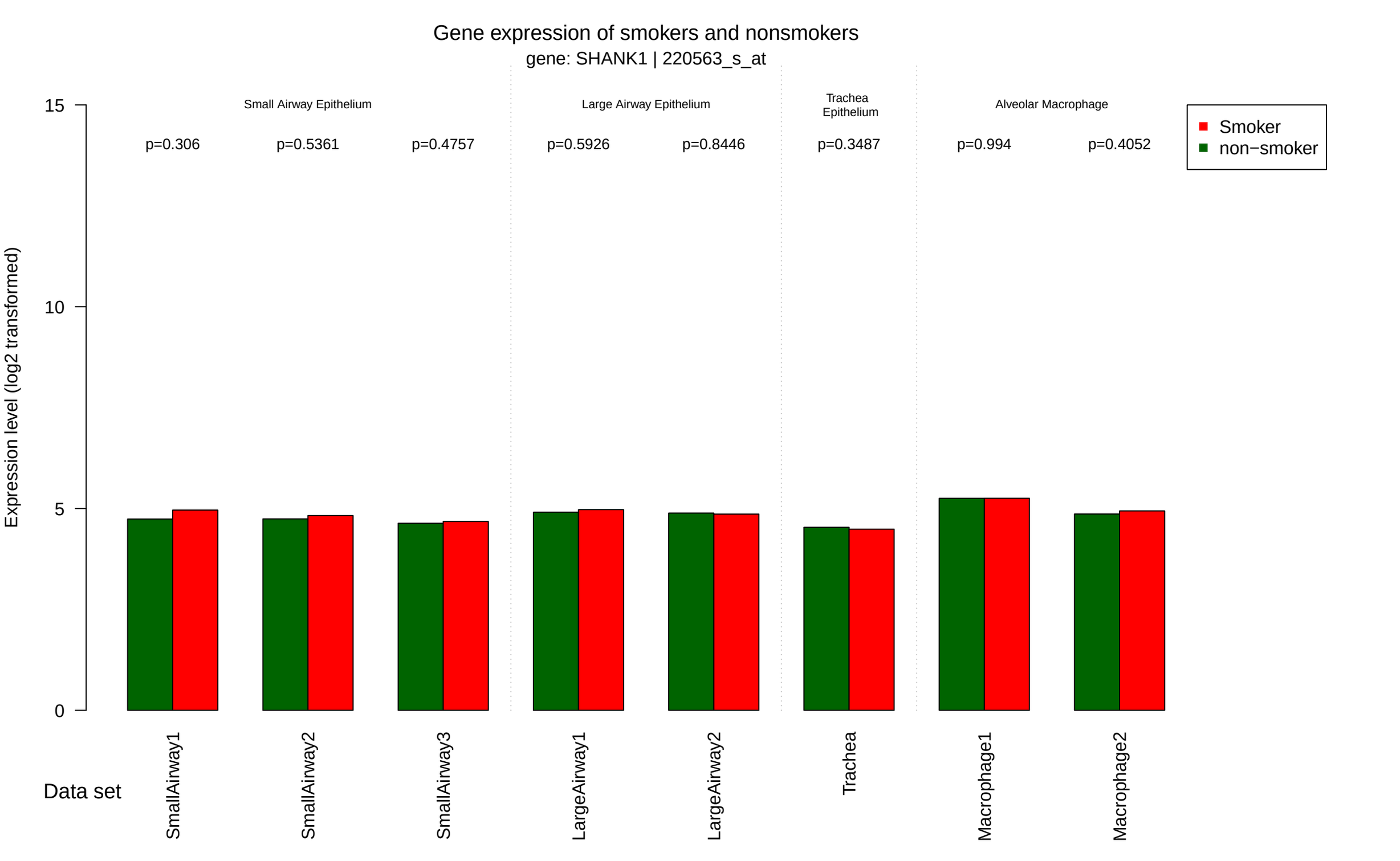
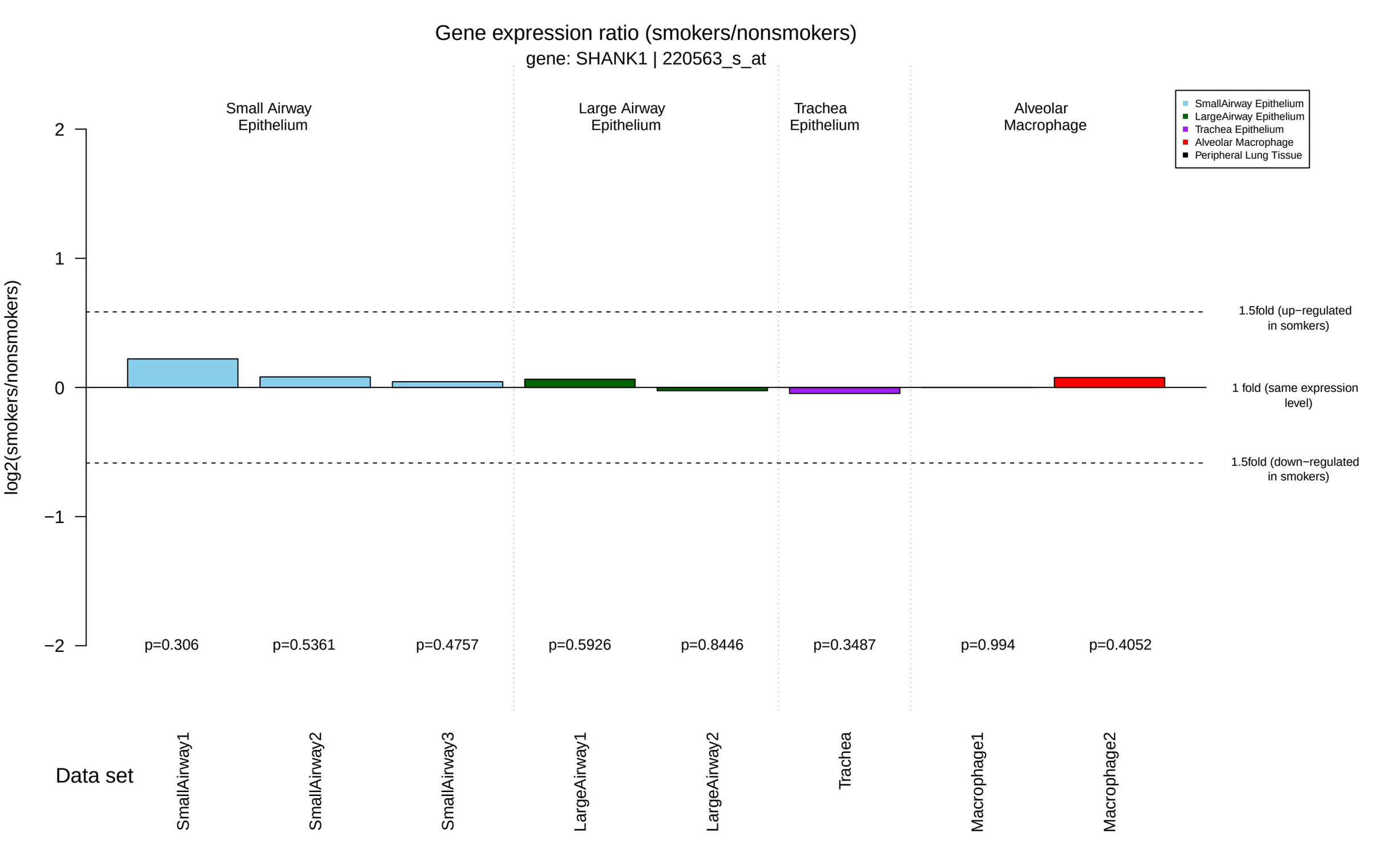
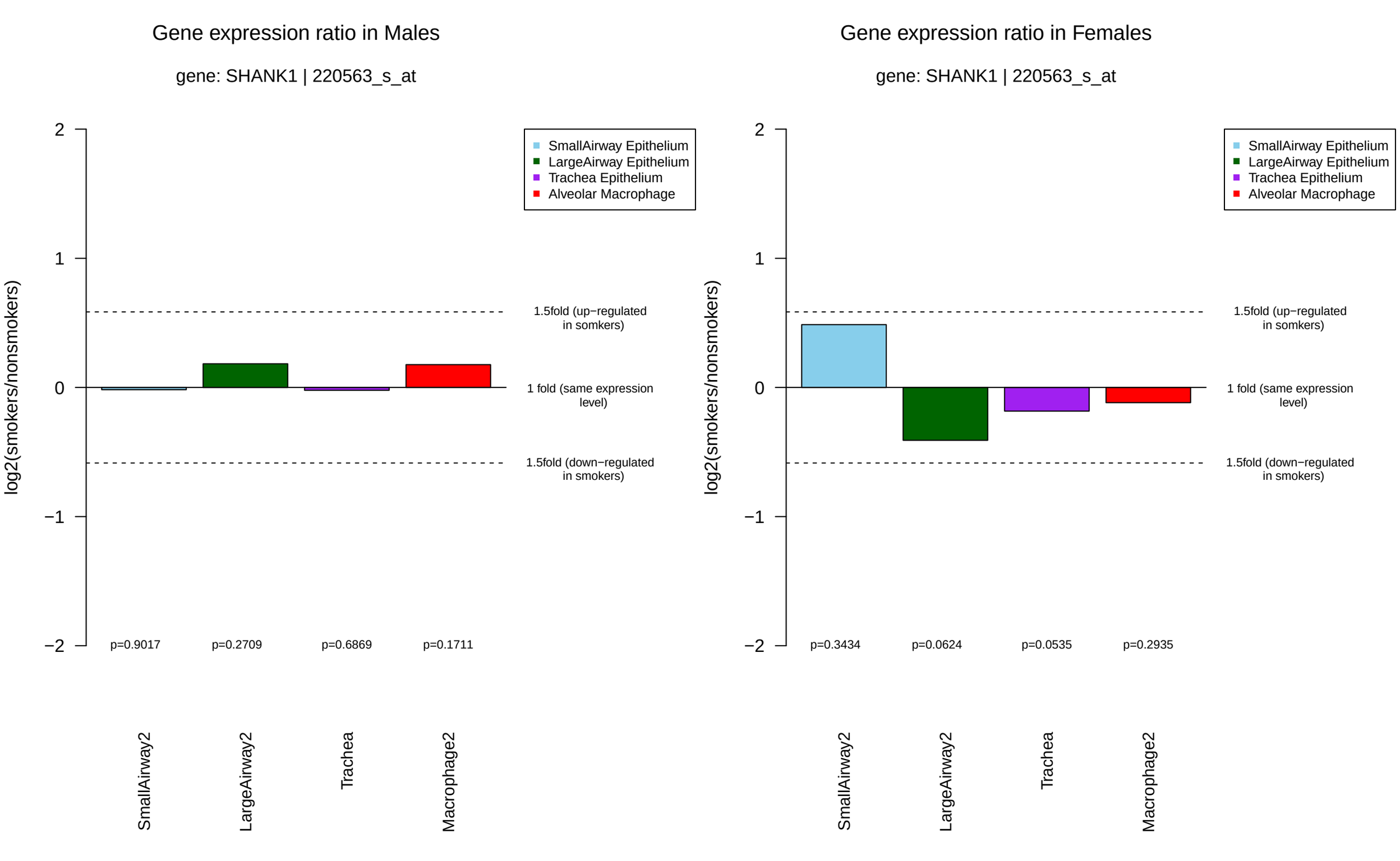
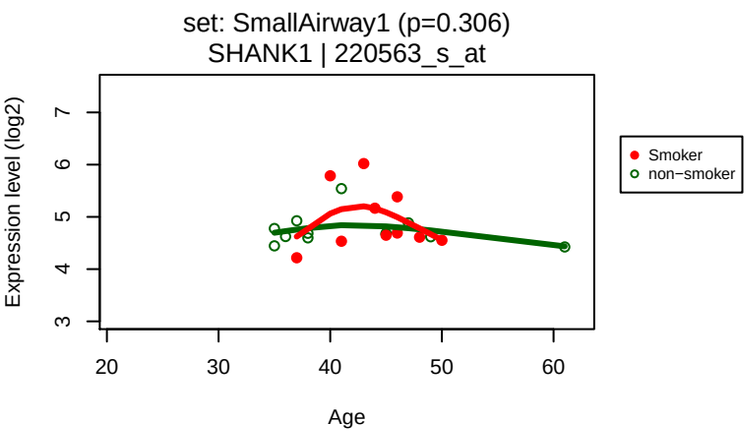
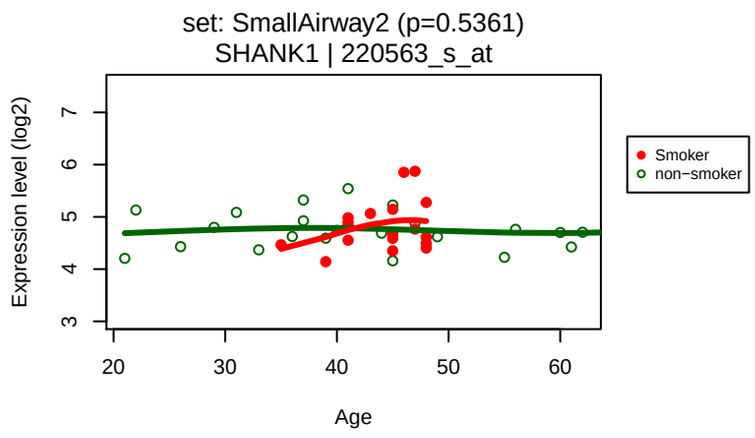
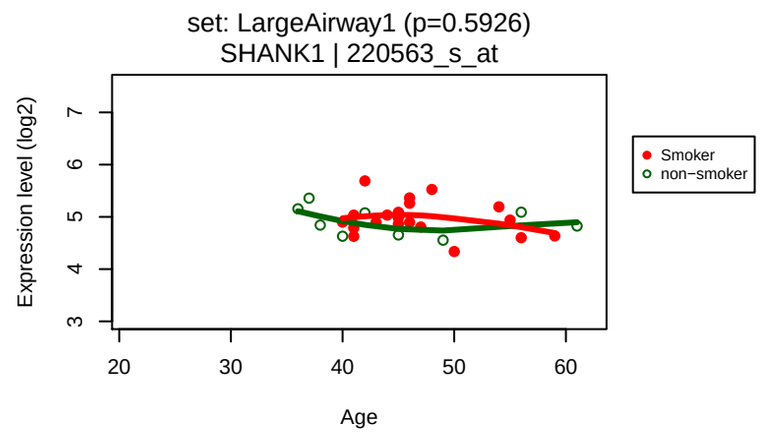

Gene expression of smokers and nonsmokers
gene: SHANK1 | 220563_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.306
p=0.5361
p=0.4757
p=0.5926
p=0.8446
p=0.3487
p=0.994
p=0.4052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

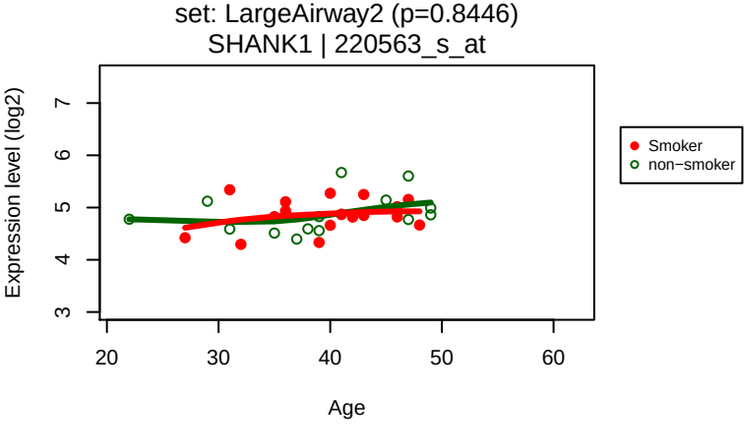

Gene expression ratio (smokers/nonsmokers)
gene: SHANK1 | 220563_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.306
p=0.5361
p=0.4757
p=0.5926
p=0.8446
p=0.3487
p=0.994
p=0.4052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

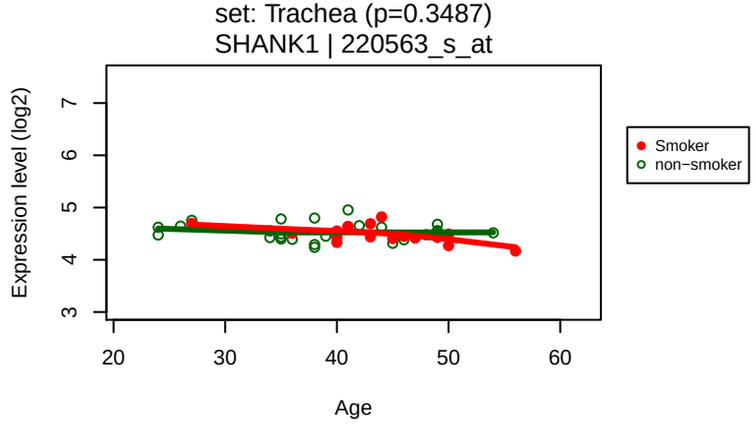

Gene expression ratio in Males
gene: SHANK1 | 220563_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9017
p=0.2709
p=0.6869
p=0.1711
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SHANK1 | 220563_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3434
p=0.0624
p=0.0535
p=0.2935
SmallAirway2
LargeAirway2
Trachea
Macrophage2

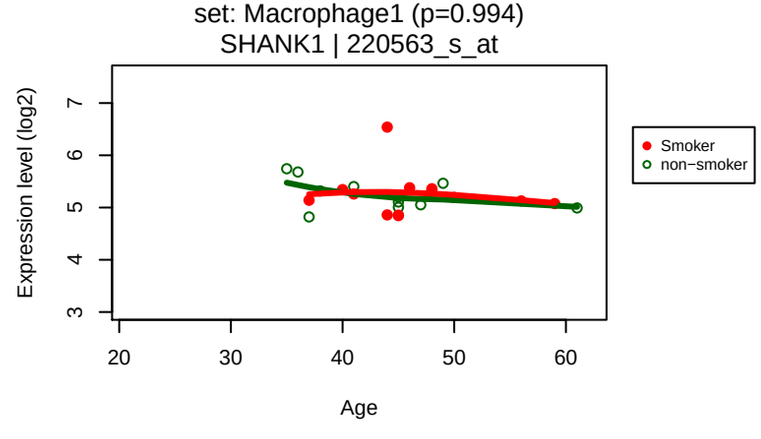

set: SmallAirway1 (p=0.306)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker

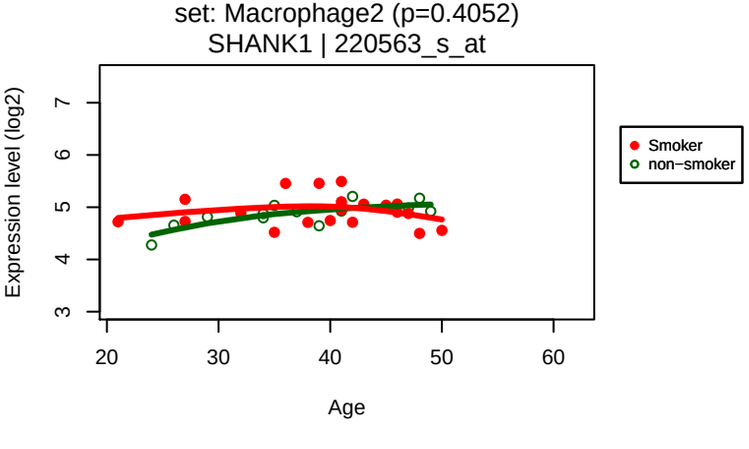

set: SmallAirway2 (p=0.5361)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker

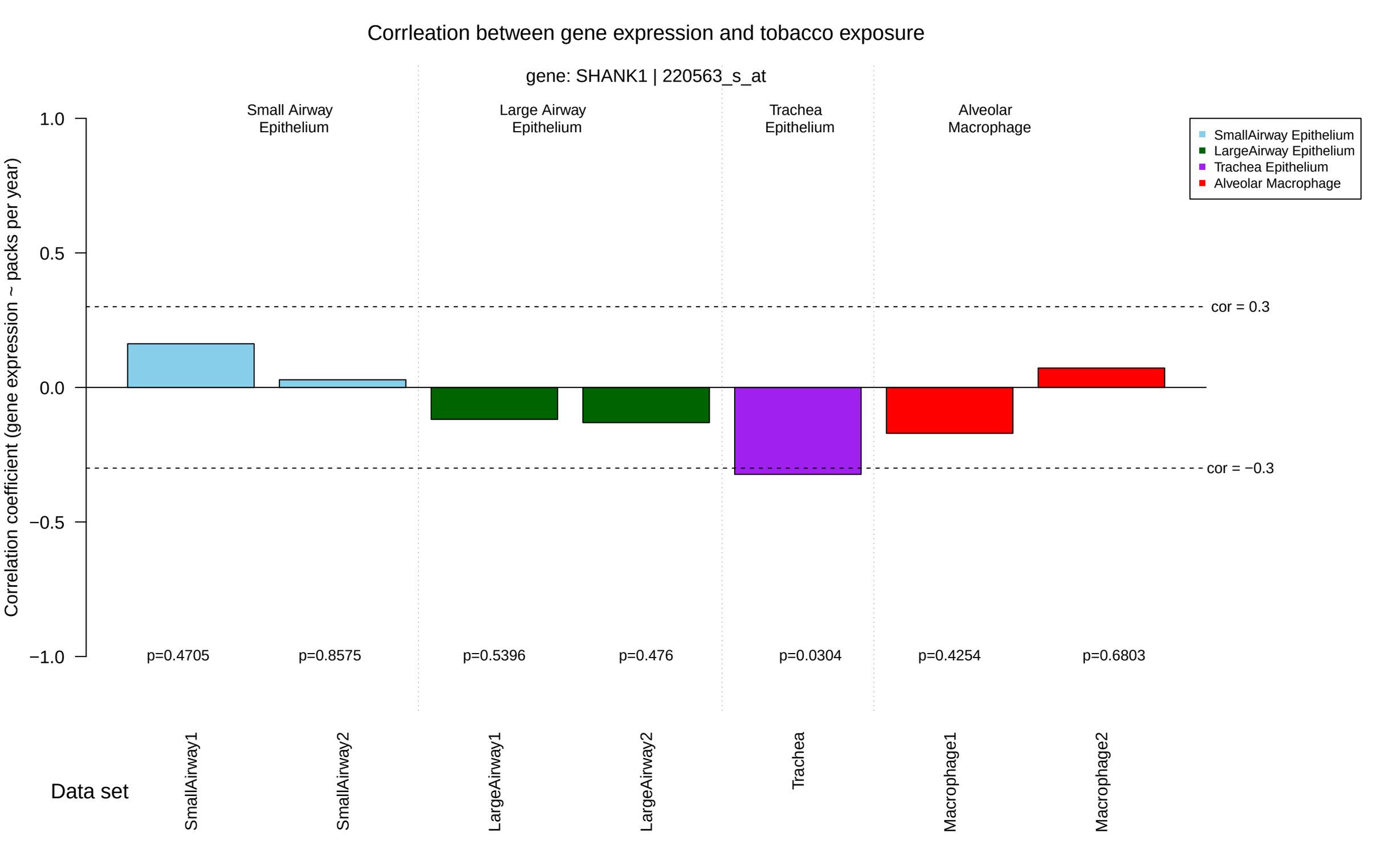

set: LargeAirway1 (p=0.5926)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8446)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3487)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.994)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4052)
SHANK1 | 220563_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SHANK1 | 220563_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4705
p=0.8575
p=0.5396
p=0.476
p=0.0304
p=0.4254
p=0.6803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2